SKILL BUILDERS · UNIT 4 · SCORE 3

Angle Relationships: Practice Set

NC.8.G.5

ANSWER KEY

Do one problem at a time. Show your steps. Circle the letter or write the number.

1 Score 3 NC.8.G.5

Find the **complement** of 28°.

Answer: **62°**

1. `90 - 28 = 62°`

2 Score 3 NC.8.G.5

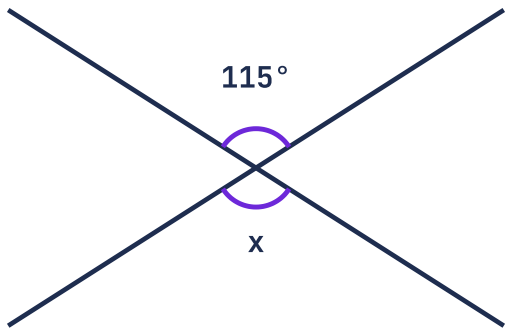
Find the **supplement** of 73°.

Answer: **107°**

1. `180 - 73 = 107°`

3 Score 3 NC.8.G.5

Two lines cross. Find **x**.

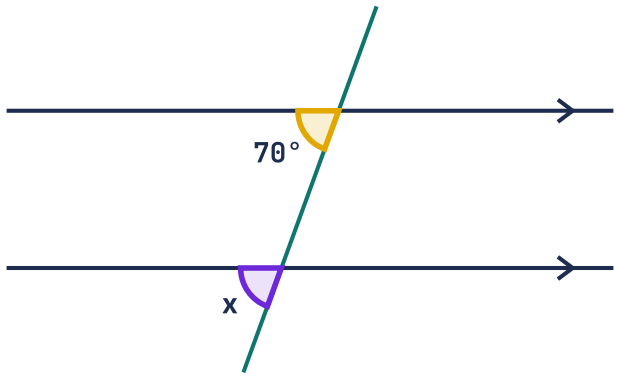


Answer: **115°**

1. `vertical angles → equal`
2. `x = 115°`

4 Score 3 NC.8.G.5

The lines are parallel. Find **x**.

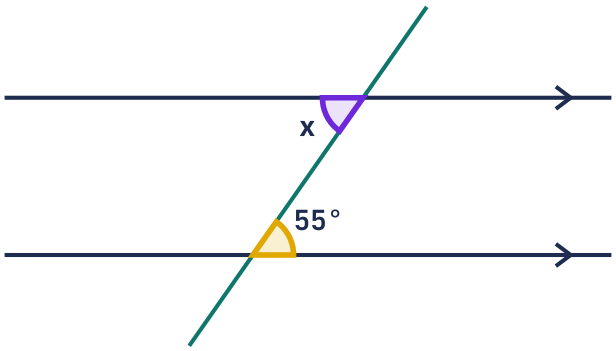


Answer: **70°**

1. `corresponding → equal`
2. `x = 70°`

5 Score 3 NC.8.G.5

The lines are parallel. Find **x**.

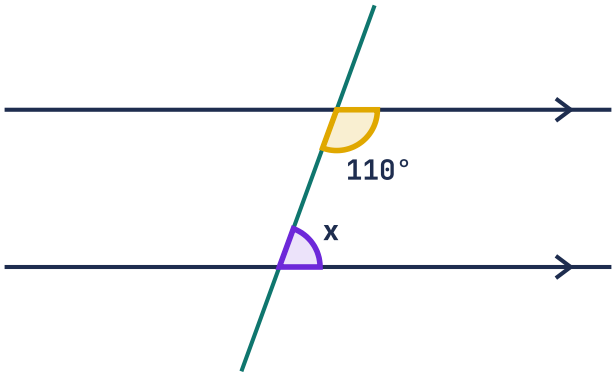


Answer: **55°**

1. alternate interior → equal
2. x = 55°

6 Score 3 NC.8.G.5

The lines are parallel. Find **x**.

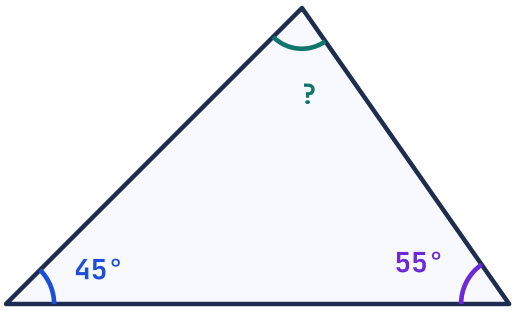


Answer: **70°**

1. same-side interior → add to 180°
2. x = 180 - 110 = 70°

7 Score 3 NC.8.G.5

A triangle has angles 45° and 55°. Find the third angle.

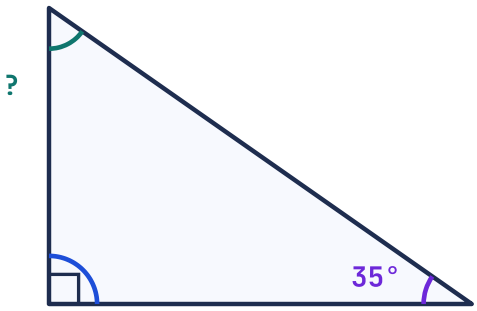


Answer: **80°**

1. 45 + 55 = 100
2. 180 - 100 = 80°

8 Score 3 NC.8.G.5

A right triangle has one angle of 35°. Find the other small angle.



Answer: **55°**

1. 90 + 35 = 125
2. 180 - 125 = 55°

9 Score 3 NC.8.G.5

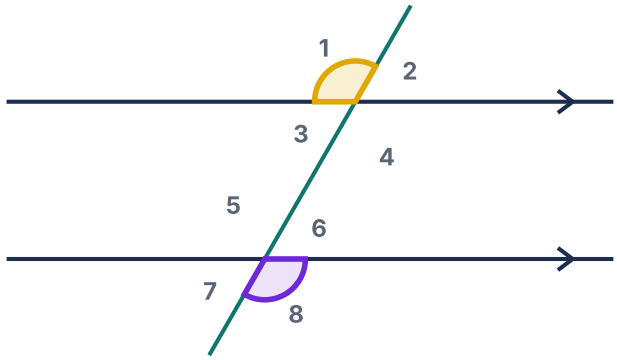
The far inside angles of a triangle are 40° and 65°. Find the exterior angle.

Answer: **105°**

1 `40 + 65 = 105°`

10 Score 3 NC.8.G.5

What kind of pair are ∠1 and ∠8?



Ⓐ Alternate interior angles

Ⓑ **Alternate exterior angles**

Ⓒ Corresponding angles

Ⓓ Same-side interior angles

11 Score 3 NC.8.G.5

Find the mistake. Ava wrote: "The supplement of 40° is 50°." What went wrong?

Ⓐ **Ava used 90. Supplementary angles add to 180, so the answer is 140°.**

Ⓑ Ava is right: 40 + 50 = 90.

Ⓒ It should be 40° because the angles are equal.

Ⓓ It should be 320° because 360 − 40 = 320.